(Please write your Enrollment Number)

Enrollment No. 025810022023

MID-TERM EXAMINATION

(Course Name B.Tech. CSE/ECE/Reappear) (Semester 1st)

(March, 2024) OFF LINE mode

Subject Code: BMA 110	Subject: Engineering Mechanics
Time : 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1 (2.5*4)

(a) Draw the free body diagram of the sphere P supported as shown in Figure 1 correctly showing the directions of the forces acting on P and the angles between them.

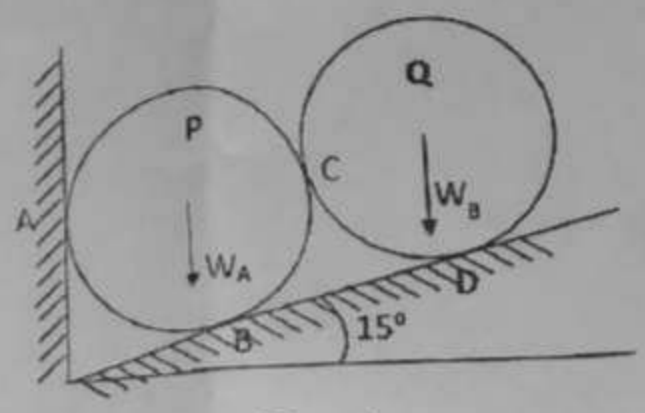


Figure 1

(b) Define angle of friction with the help of a diagram.

(c) Find the coordinates of the centroid of the shaded area shown in Figure 2.

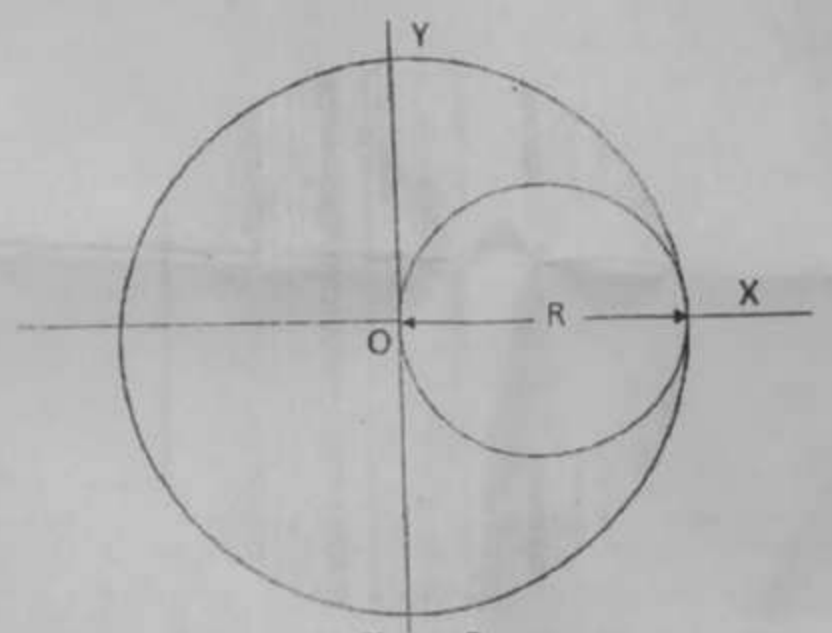


Figure 2

(d) Use proper formula to find out whether the truss shown in Figure 3 is perfect or imperfect.

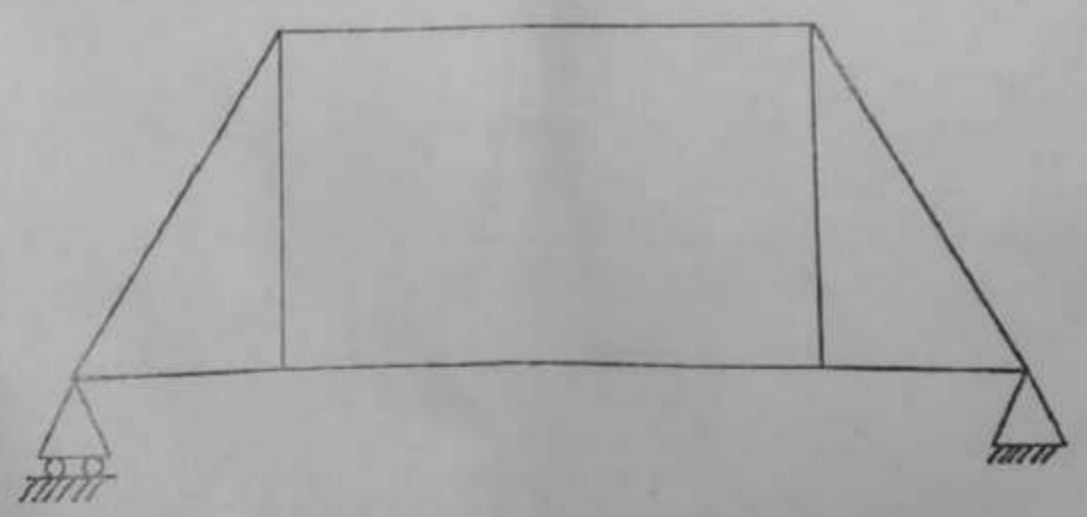


Figure 3

Q2 (Attempt any Two Parts) UNIT-1 (5,5)

(a) A uniform rod AB of 1 m length and 1 kg mass is suspended from its ends as shown in Figure 4. What amount of tensions would be induced in the strings when a mass of 5 kg is suspended from point C on the rod?

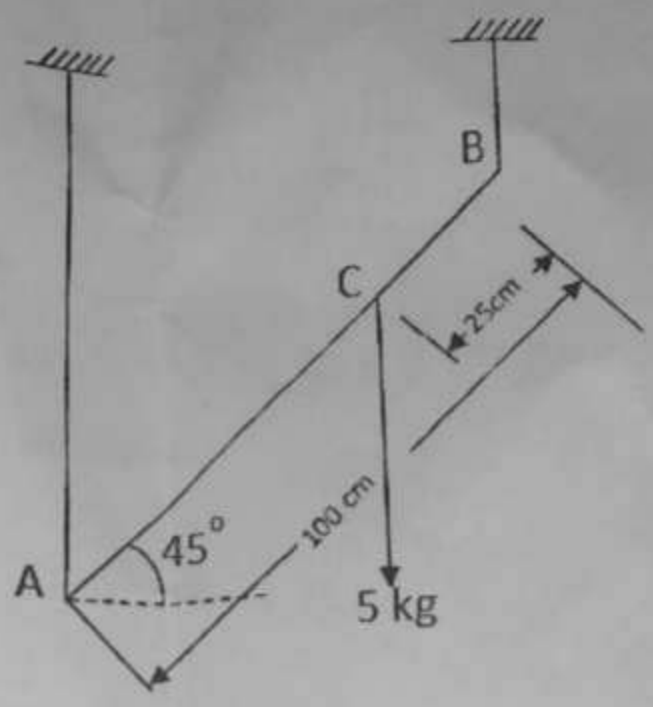


Figure 4

(b) Two like parallel forces P and Q act on a rigid body at points A and B respectively. The perpendicular distance between the lines of action of the forces is l. If P and Q are interchanged in position, their resultant gets displaced through a distance d along AB. Obtain the expression for d in terms of P, Q and l.

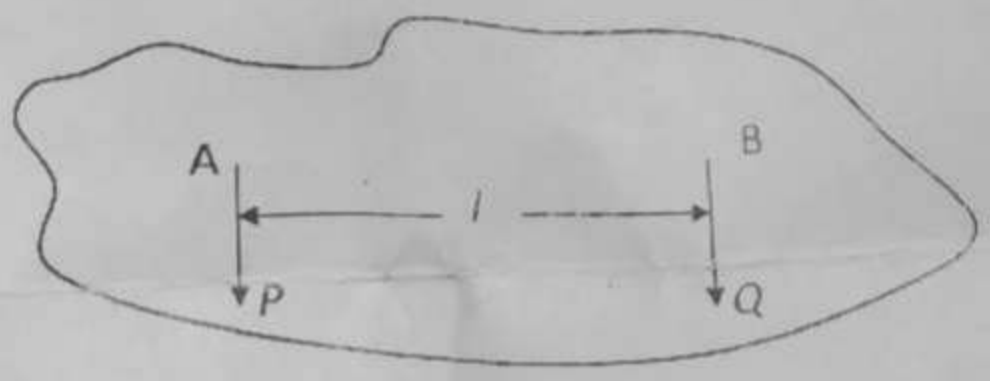


Figure 5

(c) A uniform ladder of weight 30 N and length 13 m is placed against a smooth and frictionless vertical wall with its lower end 10 m away from the wall. In this position, the ladder is just to slip. Determine: (i) the coefficient of friction between the ladder and the floor and (ii) frictional force acting on the ladder at the point of contact between the ladder and the floor.

Q3	(Attempt any Two Parts) UNIT-2	(5,5)
	(a) Derive the expression for the distance of the centroid of a semicircular lamina of radius R from its base.	
	(b) Derive the expression for the moment of Inertia I_{xx} of a semicircular lamina of radius R about its x-centroidal axis.	
	(c) Determine the support reactions and the forces in the members AB, AE and BE of the truss loaded as shown in Figure 6. Use method of joints.	

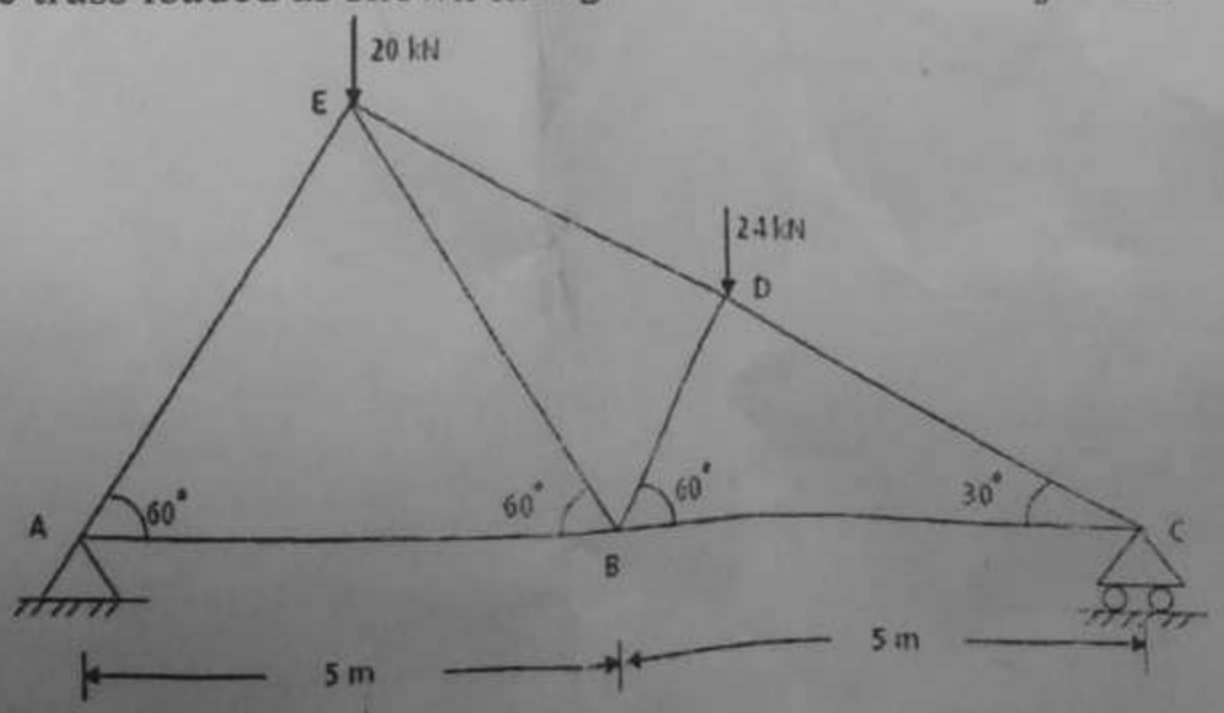


Figure 6